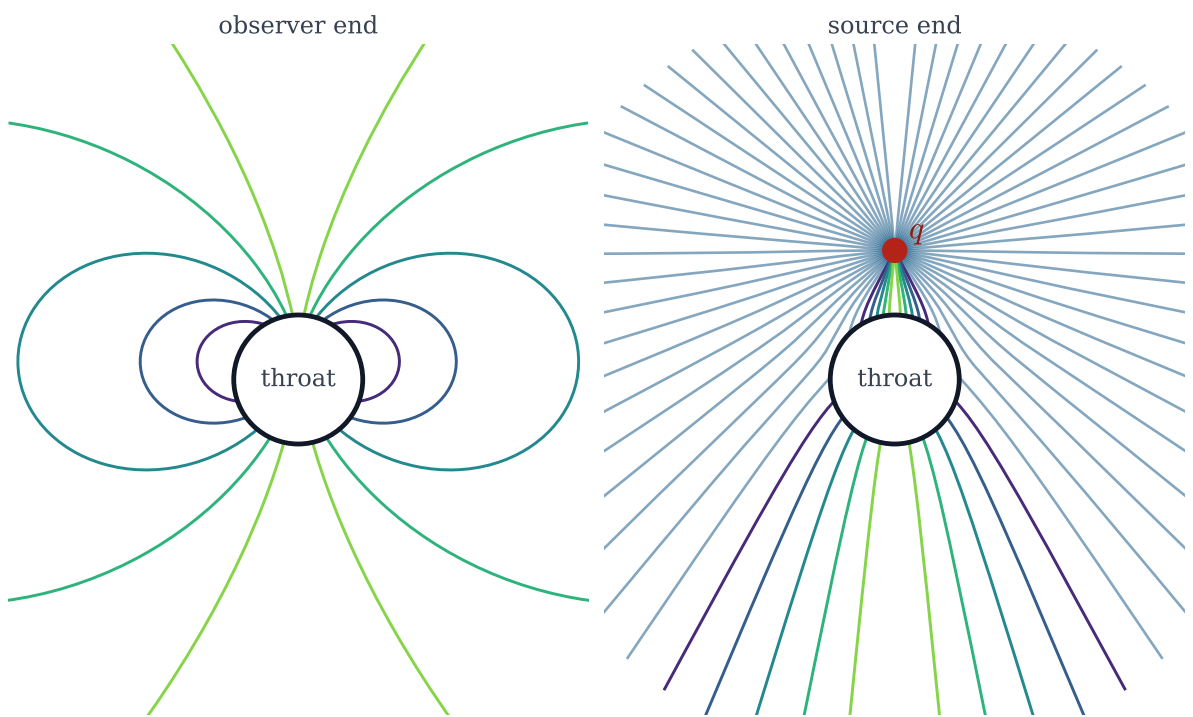
observer end
source end
throat
throat
q